

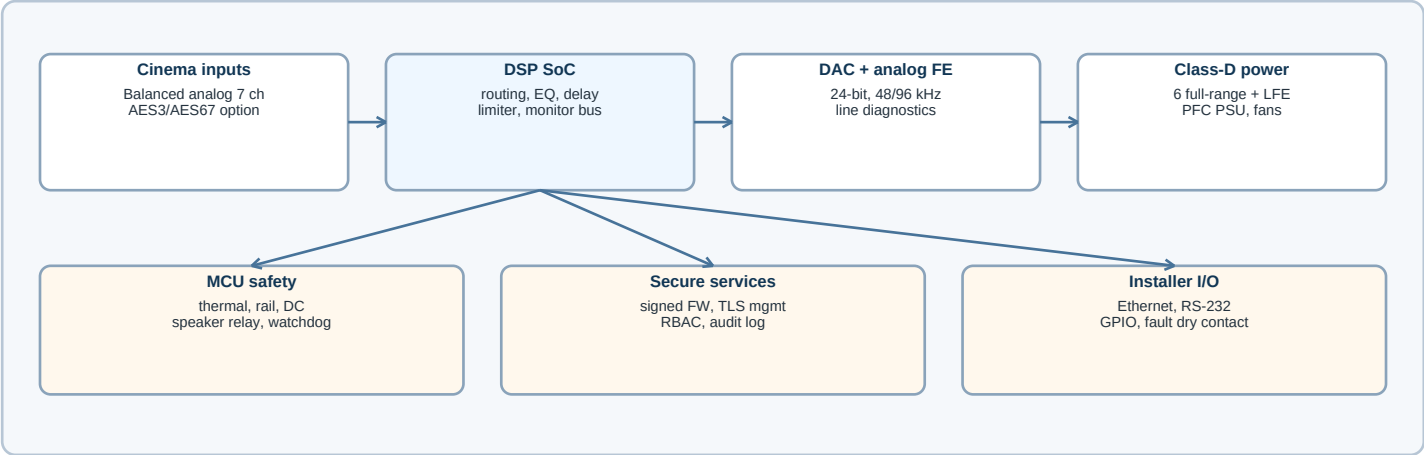
Embedded Specification - Cinema Dolby 6+1 Compatible Amplifier

Scope: seven-channel cinema power amplifier with embedded audio processing, safety supervision and secure network management. The design targets a 6 full-range + 1 LFE deployment while remaining installable behind modern cinema processors that commonly expose 7.1 or 16-channel outputs.

1. System context and market alignment

In commercial cinema, decoding, format handling and theatre routing are normally performed by an external cinema processor; current Dolby CP950-class products provide 5.1/7.1 support, 16-channel analog output and network/digital options such as AES67 or BLU Link. Therefore the amplifier shall accept line-level or network audio feeds and provide protected, monitored power amplification rather than performing proprietary Dolby decoding.

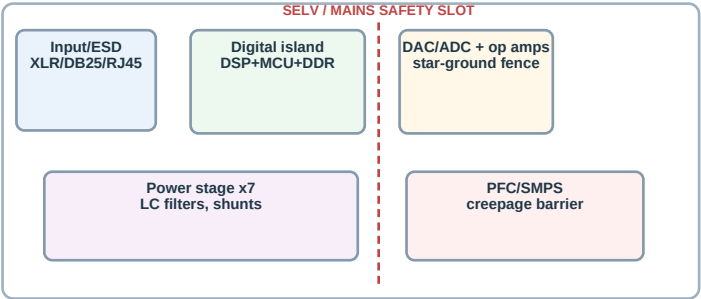
The 6+1 configuration shall map six full-range channels plus one LFE/subwoofer channel. The hardware shall be scalable to 8 amplified channels and 16 logical input channels because market products frequently use 4- or 8-channel networked cinema amplifiers and 16-output cinema processors.



2. Embedded hardware specification

PCB architecture. Four-layer minimum; six-layer preferred. Separate digital ground, low-noise analog island and high-current power return tied at controlled star points. Maintain creepage/clearance between SELV control circuits and mains/PFC areas. Place current shunts, temperature sensors and speaker relay feedback close to each output channel.

Channel set. CH1-L, CH2-C, CH3-R, CH4-Ls, CH5-Rs, CH6-Cs/rear surround, CH7-LFE. Optional CH8 spare/backup. Inputs: balanced analog on DB25/XLR or AES3/AES67 daughtercard. Outputs: low-impedance loudspeaker terminals, bridge mode disabled by default for LFE unless certified during commissioning.



Item	Minimum requirement
DSP/MCU	Audio DSP with deterministic frames; safety MCU independent from DSP reset domain
Converters	24-bit DAC path, 48/96 kHz; per-channel mute and pop suppression
Power	Class-D or high-efficiency topology; PFC SMPS; inrush limiting; fan PWM
Sensing	Rail V/I, output DC, load impedance estimate, heatsink, inlet/outlet air, fan tach
Service I/O	Ethernet, RS-232/USB service, GPIO, dry-contact fault output

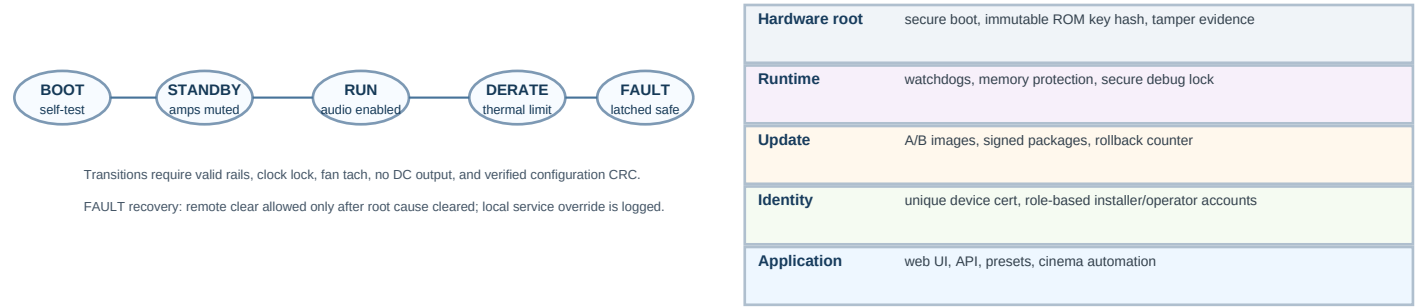
3. Software and signal-processing needs

Subsystem	Specification
Audio graph	Fixed 7-channel graph: input trim, polarity, routing, high/low-pass crossovers, parametric EQ, delay, limiter, LFE low-pass, monitor tap. Presets are versioned and checksum-protected.
Real-time control	No dynamic memory allocation in the audio callback. Configuration changes use double-buffered commit at frame boundary; no audible discontinuity above -90 dBFS.
Commissioning	Installer mode supports pink-noise channel ID, impedance sweep at reduced level, gain calibration, speaker preset import/export and locked auditorium profile.
Telemetry	1 s health frame: temperature, rail voltage, output current, limiter activity, fault history, firmware version, runtime hours; SNMP/syslog or REST/MQTT gateway optional.
Update	Signed firmware, A/B image bank, power-fail safe installation, rollback after failed boot, calibration data never overwritten by application update.

4. Performance constraints

Parameter	Target / acceptance criterion
End-to-end latency	<= 20 ms analog input to speaker output for local processing path; fixed and reported to cinema automation.
Sample rate	48 kHz mandatory, 96 kHz supported for network/digital input; internal filters coefficient-stable across rates.
Noise / dynamic range	Amplifier idle noise inaudible at auditorium reference gain; SNR target >= 100 dBA referenced to rated power.
Frequency response	Full-range channels 20 Hz-20 kHz within +/-1 dB excluding configured crossovers; LFE path 20-120 Hz default.
Limiter behavior	Look-ahead or fast RMS/peak limiter; protect driver thermal limits without more than 1 dB channel-to-channel mismatch in surround groups.
Availability	Cold boot to safe standby <= 10 s; warm fault isolation <= 500 ms; audio mute on critical fault <= 20 ms.

5. Functional safety, electrical safety and security



Safety requirements. Independent hardware mute path shall override DSP software. Critical events include output DC, over-current, over-temperature, rail undervoltage/overvoltage, fan stall, clock loss and configuration CRC failure. Fault state shall be latched where speaker damage is possible. Design validation shall include IEC/EN 62368-1 style hazard-based safety review, EMC pre-compliance, surge/burst immunity and abnormal-load tests.

Security requirements. Management services are disabled until commissioned. Default credentials are prohibited. Remote access requires TLS, unique certificates, role-based accounts, lockout after repeated failures and complete audit logging. Debug interfaces are locked in production; service unlock requires a time-limited signed token. The device must continue audio or fail safe during network attacks; network management must never starve the audio thread.

6. Market sanity check and verification matrix

Market observation	Design implication
Dolby CP950/CP950A processors expose 5.1/7.1 workflows and 16 analog or digital outputs, with EQ/crossover functionality and AES67/BLU Link options.	This unit is specified as an amplifier-side endpoint. It does not implement proprietary cinema decoding; it accepts already-rendered channel feeds and preserves deterministic gain, delay and protection.
Q-SYS DPA-Q cinema amplifiers are publicized as 4- and 8-channel network amplifiers with GPIO, Q-SYS monitoring and energy-management behavior.	A 7-channel amplifier is commercially plausible when implemented as an 8-channel board with one spare channel, GPIO, remote health reporting and installer-oriented monitoring.
Crown/JBL DSi 2.0 cinema amplifiers are positioned for cinema loudspeakers with onboard DSP and Dante/AES67-capable variants.	The embedded scope shall include DSP presets, limiters, network input options and fleet-management functions, while loudspeaker presets remain locked after commissioning.
Test area	Representative verification
Audio	Channel mapping, pink-noise ID, EQ coefficient accuracy, latency measurement, overload recovery, LFE crossover response.
Power/safety	Shorted load, open load, low impedance, thermal chamber, fan blocked, mains dip/surge, emergency mute, relay weld detection.
Software	Unit tests for routing and state machine, HIL watchdog tests, secure update interruption, corrupted preset rollback.
Cybersecurity	Credential scan, TLS configuration, port scan, fuzz management API, unsigned firmware rejection, audit-log integrity.

Source notes. Dolby CP950/CP950A public product sheets and manuals; Q-SYS DPA-Q public specifications and hardware overview; Crown/JBL DSi 2.0 public product information. These sources support the architecture choice and prevent over-specifying Dolby decoding inside the amplifier.